

Hamza Bouktitiya

Artificial Intelligence Engineer

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LinkedIn • GitHub • Portfolio

AI engineer building production LLM systems (RAG, agents, APIs) with FastAPI, LangChain/LangGraph, and Docker. Experience spanning NLP, computer vision, and full-stack deployment.

Education

- **Master in Artificial Intelligence and Virtual Reality** *2022 – 2024*
  Ibn Tofail University, Kenitra, Morocco

- **Bachelor in Mathematical and Computer Science** *2018 – 2022*
  Ibn Tofail University, Kenitra, Morocco

- **Baccalaureate in Mathematical Sciences** *2017 – 2018*
  Sidi Aissa High School, Souk El Arbaa Du Gharb

Professional Experience

- **AI Developer** *April 2025 – October 2026*
  OBYSTECH SOLUTIONS, Rabat
  Advise enterprises on AI adoption (ML, deep learning, NLP) and ERP integrations such as Odoo; shipped a multi-department AI agent platform solo.
  - Identify AI use cases across CRM, sales, inventory, and HR; prototype NLP/ML solutions wired into Odoo and enterprise workflows.
  - Deliver NLP assistants that let users trigger ERP actions in natural language across Odoo modules (CRM, sales, inventory, HR).
  - Built Next.js multi-chat UI (RAG toggle, team workspaces) and FastAPI/LangGraph RAG pipeline (HF embeddings, hybrid search, MCP tools, PostgreSQL memory).
  - Secured LLM APIs (chat, RAG, image generation) with typed schemas, JWT/OAuth2, and role-based access for users and collections.
  - Deployed multi-service Docker Compose on VPS (reverse proxy, SSL, zero-downtime).
  - **Stack:** Next.js, FastAPI, LangChain, LangGraph, MCP, Ollama, Odoo, PyTorch, NLP, PostgreSQL, ChromaDB, Docker, JWT/OAuth2.
- **AI Intern** *February 2024 – August 2024*
  Verve Technologie, Casablanca
  - Built an LSTM forecasting system for energy consumption prediction.
  - Exposed Stable Diffusion 1.5 image-generation endpoints and a RAG pipeline over Wikipedia/PDF; FastAPI with JWT auth.
  - Designed REST endpoints and user management for continuous LLM/image interactions.
  - **Stack:** LSTM, LangChain, FastAPI, Stable Diffusion, JWT.
- **Web Development Intern** *March 2022 – June 2022*
  CS Department, Ibn Tofail University, Kenitra
  Developed a gym management web app for clients and administrators (PHP, JavaScript, Bootstrap, MySQL).

Technical Skills

AI / ML	PyTorch, TensorFlow, LangChain, LangGraph, CrewAI, RAG, LLMs, NLP, Computer Vision, NeRF
Back-end	FastAPI, Django, Python, REST APIs, PostgreSQL, MySQL, JWT/OAuth2
Front-end	Next.js, React, TypeScript, Tailwind CSS, HTML/CSS, Streamlit
MLOps / Ops	Docker, Docker Compose, Nginx, VPS deployment, Git/GitHub
Data	Pandas, NumPy, Power BI, TF-IDF, classical ML (SVM, KNN)

Selected Projects

- **Centre Souifi** (2026, ongoing) — Multilingual marketing site (AR/FR/EN, RTL) for an evening tutoring centre; static HTML/CSS/JS on GitHub Pages.
- **Intelligent Assessment System** (2026) — LLM/RAG exam generation, evaluation, and feedback (BLEU, ROUGE, BERTScore); FastAPI, Streamlit, LangChain, Docker.
- **Computer Vision Playground** (2025) — Streamlit/Docker: MTCNN, LBPH, YOLOv8, MediaPipe pose.
- **License Plate Recognition** (2023) — HOG + KNN/SVM; **98.9%** test accuracy.

Languages & Activities

Languages: Arabic (Native) • English (C1) • French (B2)

Activities: DyslexAI Hackathon — Top 3 (Al Akhawayn University) • ThinkAI Hackathon (1337 Coding School) •